# Full amyloid status cerebellum - Insight46 Phase 1

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## Document details

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|---|---|
| **Categories of variables** | florbetapir-PET SUVRs with cortical composite target region and whole cerebellum reference region |
| **Summary of work undertaken** | Composite SUVR values (compcortgm) extracted from source file, status added according to cutpoint. See below for methods. |
| **Names of source variables** | Subject, ID, Scan, Date, •, And, for, both, non-PVC, and, PVC, values:, •, frontal, parietal, precuneus, occipital, temporal, antmidcingulate, postcingulate, insula, hippocampus, caudate, thalamus, putamen, compcortgm, cortgm, subcortgm |
| **Output variables** | amyld_scndte_i46p1, •, suvr_crb_i46p1, •, suvrpvc_crb_i46p1, •, amyld_imp_i46p1, •, amyld_com_i46p1, •, amyld_stscrb_i46p1, •, amyld_pvcst_i46p1 |

## Methods:

**Adapted from:** Lane, C. A., Parker, T. D., Cash, D. M., Macpherson, K., Donnachie, E., Murray-Smith, H., ... & Brown, D. (2017). Study protocol: Insight 46–a neuroscience sub-study of the mrc national survey of health and development. BMC neurology, 17(1), 75.

## Image Acquisition:

Scanning was performed on a Biograph mMR 3 T PET/MRI scanner (Siemens Healthcare, Erlangen). Amyloid load is assessed using the 18F amyloid PET ligand, florbetapir. After intravenous cannulation, 370 MBq florbetapir F18 (Amyvid) is injected. PET data are acquired continuously during and following injection to allow florbetapir uptake dynamics to be assessed. Final amyloid burden is assessed over a 10-min period, ~50 min after injection, with scope for the previous 10-min period to be used if longer scan periods are not tolerated.

PET data, acquired in list-mode, is reconstructed using a 3D ordered-subset expectation-maximisation algorithm with three iterations and 21 subsets, and smoothed with a 4 mm Gaussian kernel. Attenuation maps are computed from the T1-weighted and T2-weighted volumetric scans using a multi-atlas CT synthesis method [76], also known as pseudo-CT (pCT).

## Image Processing:

T1 images were parcellated using GIF v3 and then parcellations were registered to the PET images. Global standardised uptake value ratios (SUVRs) were calculated from a composite cortical target region normalised to a whole cerebellum reference region and with and without partial volume correction (PVC) using the Discrete Iterative Yang technique.

Positive or negative amyloid status was determined using a Gaussian mixture model applied to SUVRs, with the 99th percentile of the lower Gaussian as the cutoff (non-PVC > 1.0770 and PVC > 1.0306).

**PLEASE NOTE** that this pipeline has been performed for the cross-sectional analysis of SUVR only. SUVR will be computed in a longitudinally consistent fashion on both timepoints, and this may result in small changes in SUVR (<0.5% in most cases and < 2% in nearly all cases) when comparing values generated in cross-sectional pipeline with those in the longitudinal pipeline.

## Variables in this document

### 5 medium-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Sub-Studies [57] › Insight46 [699] › Brain PET [6034] › Amyloid data [60341] › Cross-sectional beta-amyloid status (phase 1) [603411]** | | |
| amyld_imp_i46p1 | Imputed amyloid-beta status with whole cerebellum reference region - Insight 46 phase 1 | prose scan |
| amyld_pvcst_i46p1 | Amyloid-beta status with whole cerebellum reference region with partial volume correction (PVC) - Insight 46 phase 1 | prose scan |
| amyld_stscrb_i46p1 | Amyloid-beta status with whole cerebellum reference region - Insight 46 phase 1 | prose scan |
| suvr_crb_i46p1 | Global standardised uptake value ratios (SUVR) with whole cerebellum reference region - Insight 46 phase 1 | prose scan |
| suvrpvc_crb_i46p1 | Global standardised uptake value ratios (SUVR) with with whole cerebrellum reference region with partial volume correction (PVC) - Insight 46 phase 1 | prose scan |